

Mapping the Wait: What Drives GP Appointment Delays Across England (2022–2024)

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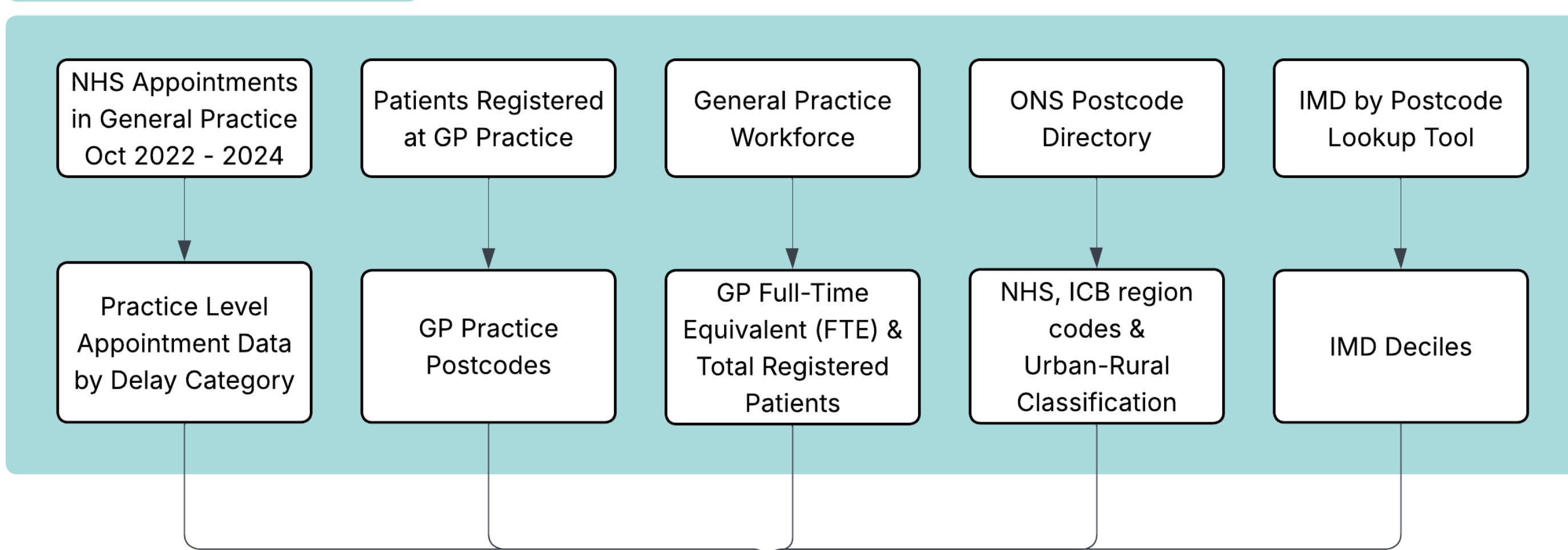
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Introduction

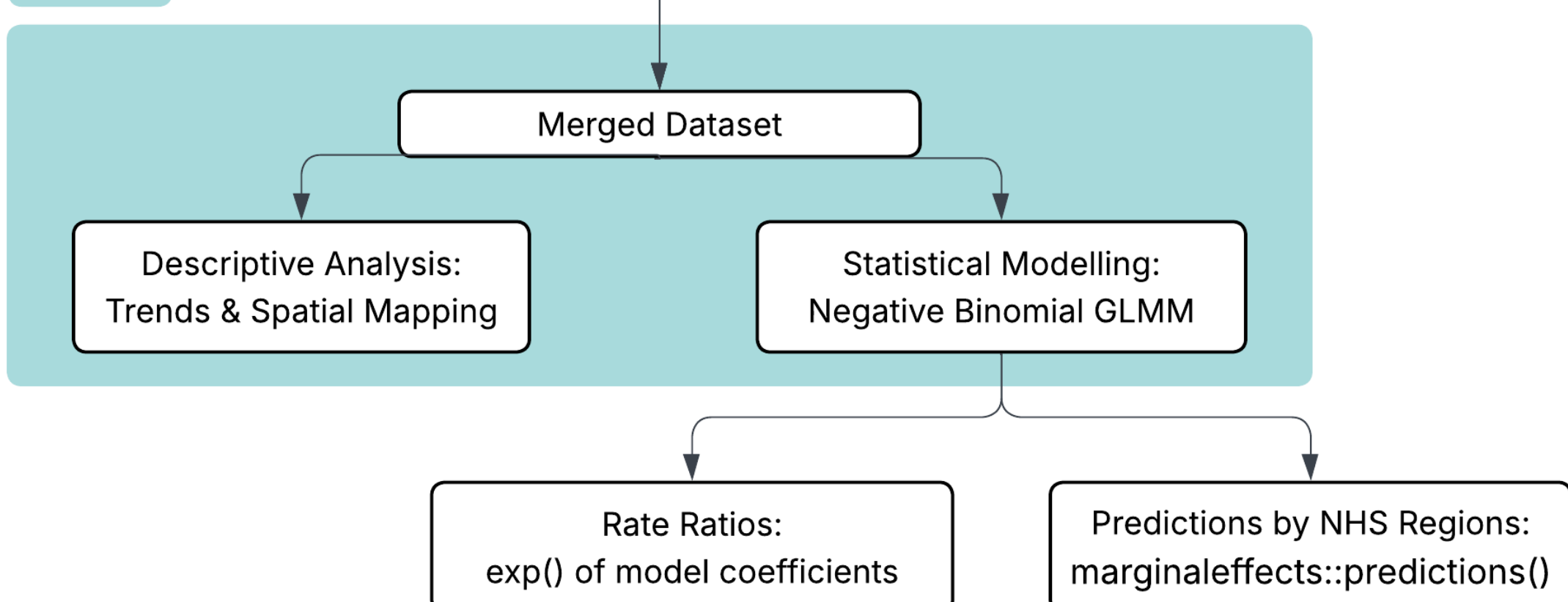
General practice (GP) forms the backbone of the NHS, delivering **over 300 million appointments annually** and serving as the **primary entry point** for patients seeking healthcare^{1,2}. Despite this crucial role, public satisfaction with GP appointment availability has declined dramatically, with **62% of patients reporting dissatisfaction** with wait times in **2024**³. These delays carry significant consequences, including **postponed diagnoses, preventable complications, and increased burden on emergency services**⁴. While previous research has focused primarily on patient outcomes and experiences, the **organizational, spatial, and structural factors** driving these delays are understudied. This study aims to analyse **how GP appointment delays evolved from October 2022 to October 2024**, and to **identify** the key regional, temporal, and socioeconomic **factors associated with long waits (>15 days from booking)**, providing evidence to inform targeted policy interventions.

Methods

Data Collection and Preparation



Analysis



Factors Associated with Long Waits for GP Appointments

Using **negative binomial regression** to analyze predictors of appointment delays, we identified several key factors:

Geographic variation

All regions outside London had higher rates of long waits: South West (86% higher), North East and Yorkshire (79%), and East of England (67%) reinforcing regional disparities.

Socioeconomic Pattern

Less deprived areas had 30% more long waits than more deprived areas, likely due to patient care-seeking behavior, service expectations and workforce distribution^{6,7}.

Urban vs. Rural

Urban practices had 6% fewer long waits than rural areas, highlighting challenges in rural staffing and access⁸.

Seasonality

Delays peaked in October (36%), with smaller spikes in September (21%) and April (17%), linked to seasonal demand⁹.

Time Trend

Long waits rose by 0.3% per month, reflecting post-COVID demand and strain on GP services, suggesting possible inefficiencies in appointment management¹⁰.

Staffing Impact

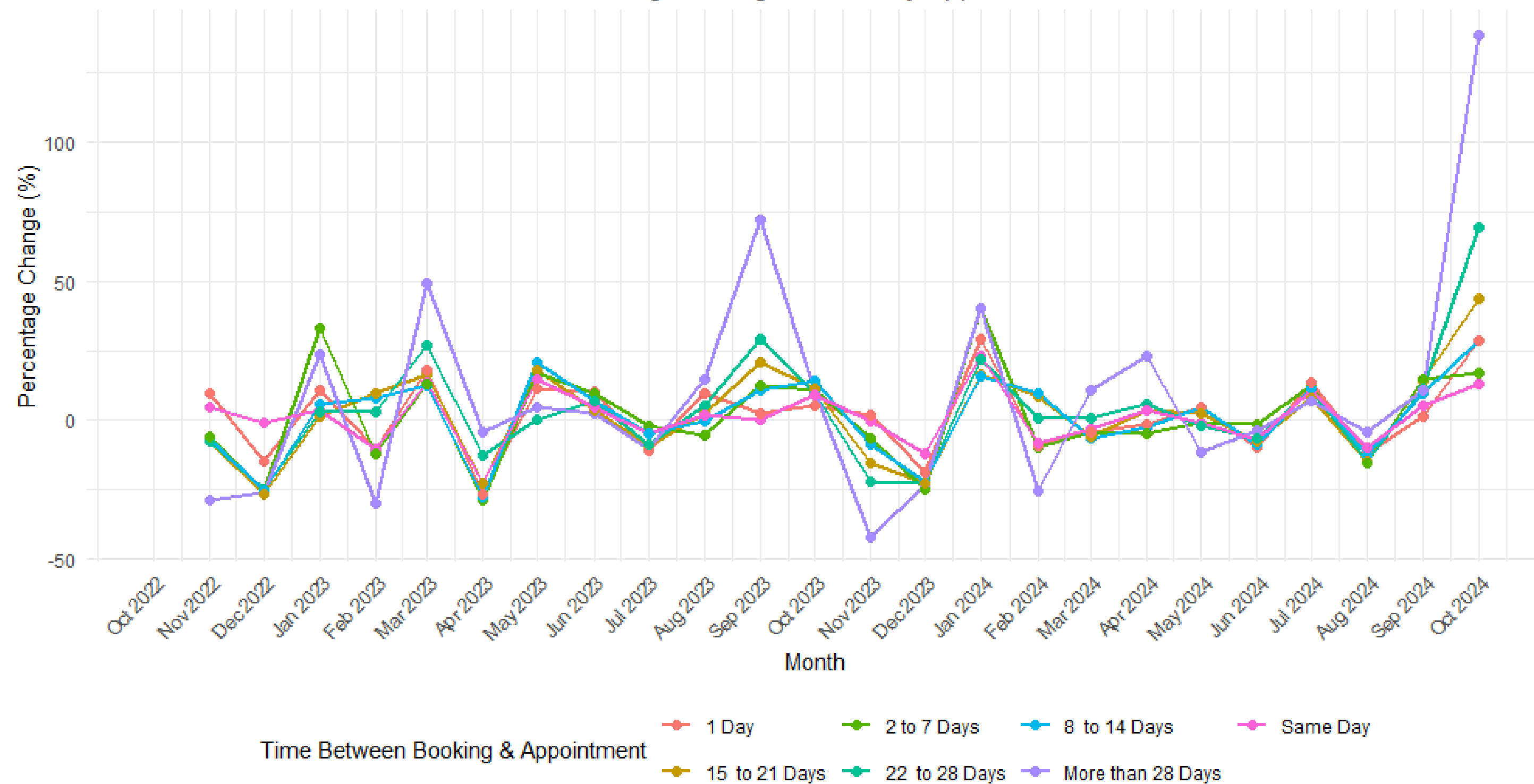
GP FTE (per 1000 patients) had minimal effect (0.03%), suggesting systemic factors such as scheduling and workflow play a greater role than workforce numbers alone^{11,12}.

Results & Discussion

Trends & Spatial Variation in Appointment Delays

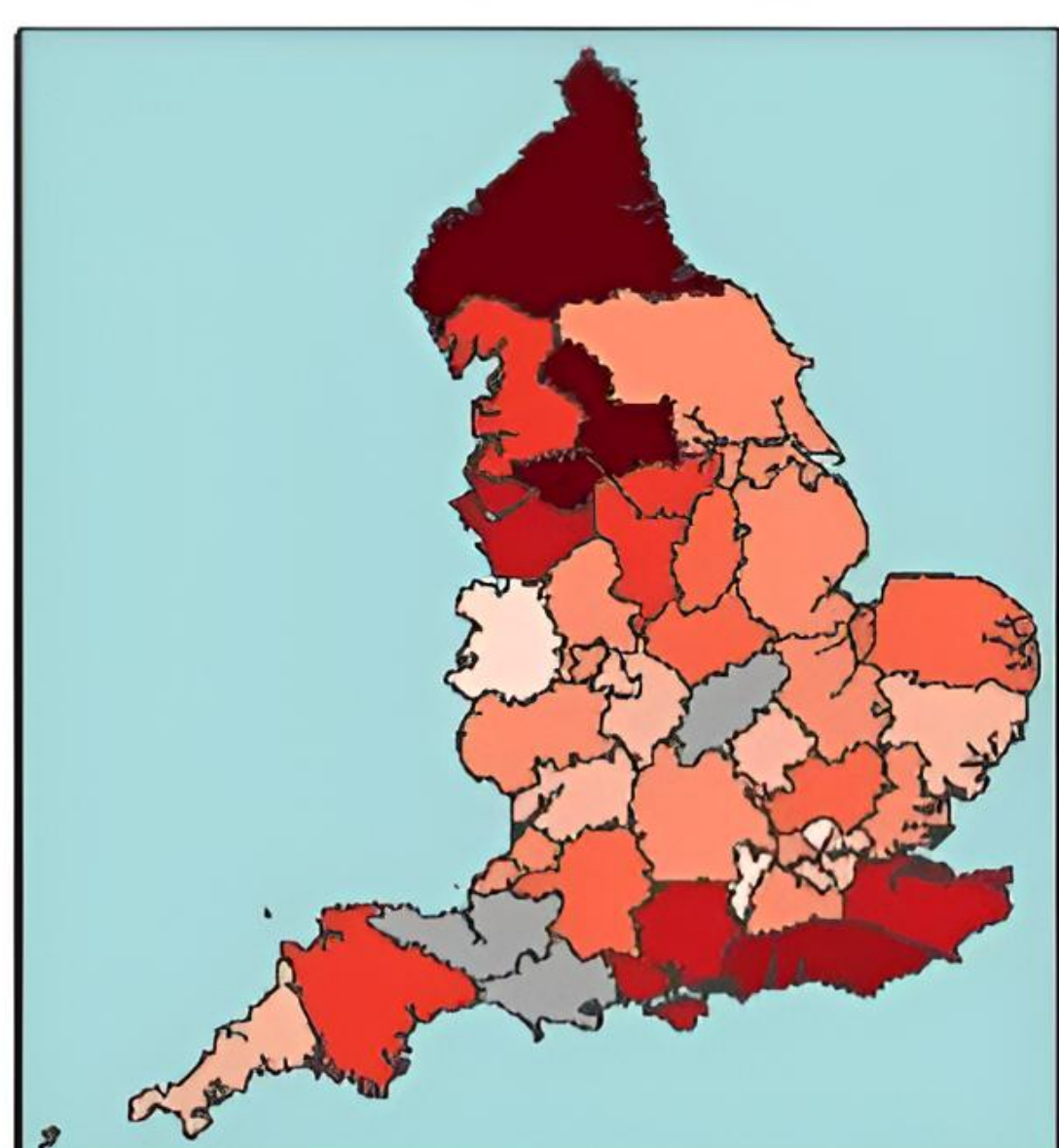
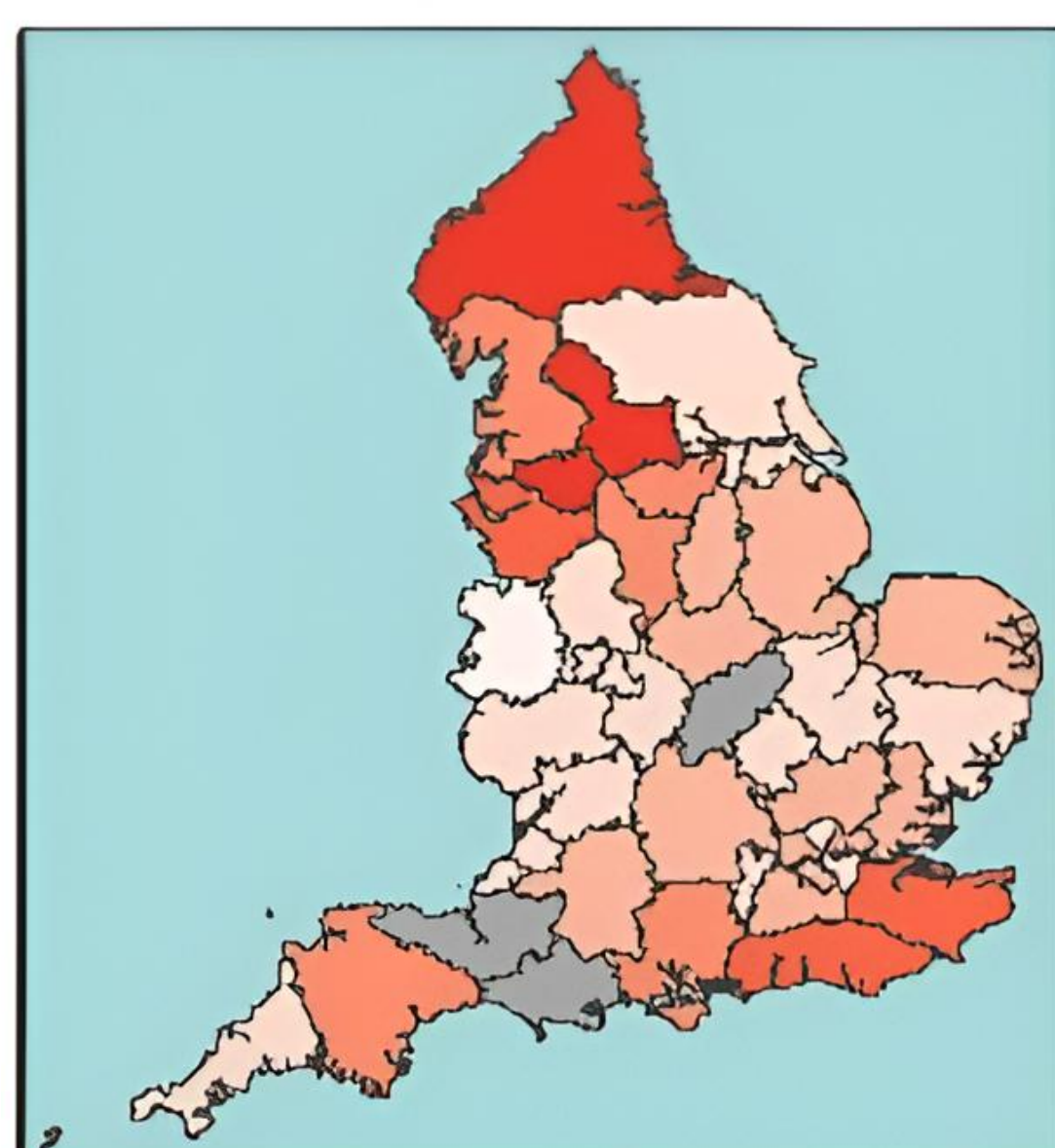
Appointments delayed >28 days **increased by over 100%** in the final study month with significant fluctuations. Geographic mapping showed that by **October 2024**, delays had expanded across England, with **northern and Midlands areas exceeding 400,000 delayed appointments**, which are substantially higher than the moderate levels (<300,000) observed in 2022. These patterns reflect varying **regional investment, digital infrastructure, and leadership styles**⁵.

Percentage Change in Monthly Appointments



October 2022

October 2024



Policy Implications

Targeted regional strategies such as **digital innovation, workforce incentives, and best-practice sharing** are needed to improve appointment equity and reduce systemic delay growth¹³.

Conclusion

- Appointment delays are rising** at an alarming rate.
- Northern and rural regions** are more heavily affected.
- Less deprived areas face longer waits**, challenging common assumptions.
- Workforce size has limited impact**; systemic and organisational factors matter more.
- Solutions lie in **smarter triage, digital infrastructure, and region-specific strategies**.
- Achieving **equitable access** demands **targeted and data-driven action**, not just increased resources.

- Stokes-Lampard H, Openshaw D. General practice at the heart of the NHS. Br J Gen Pract. 2018.
- Gajjar D et al. General practice in England. House of Commons Library; 2025.
- Taylor B et al. Public satisfaction with NHS and social care in 2024. King's Fund; 2025.
- Murray M, Berwick DM. Advanced access in primary care. JAMA. 2003.
- Medical Technology Group. Tackling regional variation in healthcare. 2022.
- Fisher R et al. Improving general practice in deprived areas. Health Foundation; 2022.
- Olah ME et al. Socioeconomic status and primary care access. CMAJ. 2013.
- Weinhold I, Gurtner S. Healthcare shortages in rural areas. Health Policy. 2014.
- Millwood S et al. Winter pressures on general practice. BJGP Open. 2021.
- Owen K et al. GP retention in the UK: a worsening crisis. BMJ Open. 2019.
- Baird B et al. Understanding pressures in general practice. King's Fund; 2016.
- Ansell D et al. Reducing wait times for primary care: a systematic review. BMC Health Serv Res. 2017.
- Office for National Statistics. National Statistics UPRN Lookup. ONS; 2024.